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Unit 3 : Session 8 : SLO 1 | Apply Exception Handling in Database Access

I. Fill in the Blanks

1. SQLException is the base checked exception for all JDBC database access errors.
2. finally the finally block (or try-with-resources) ensures database resources are closed properly.
3. AutoCloseable try-with-resources automatically closes any object that implements the AutoCloseable interface.
4. subclass superclass (inheritance) in a multi-catch block the exception types must not be related by inheritance.

II. True or False

1. TRUE SQLException provides getMessage(), getSQLState() and getErrorCode() for diagnosing database errors.
2. TRUE ResultSet and Statement must be explicitly closed in a finally block, or automatically via try-with-resources.

III. Multiple Choice

1. B catch(SQLException e) is the correct syntax for catching database-related exceptions.
2. C getSQLState() returns a standardized five-character SQL error code string.

IV. Programming

1. Identify the output: A The connection fails, the SQLException is caught, and the program prints "Error: " followed by the driver-supplied message (e.g. invalid URL / connection refused).
2. Identify the error: B executeQuery() throws SQLException, not IOException; the catch clause must declare SQLException (or Exception), otherwise the program does not compile.